PASS Data Community Summit

Host redgate

AI in the SQL World Tools We Have & Tips for Using Them

LEARNING PATHWAY · SESSION 2 OF 3

Adapting for AI Readiness as a Data Professional

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I run database solutions at GMO LLC and I'm a Microsoft Data Platform MVP. I write at sqlworldwide.com and spend my days on real production SQL Server — not lab demos.

Everything in this session runs on my laptop and my Azure subscription. Slides, scripts, and setup notes are linked at the end.

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Disclaimer



All information is as of October 19, 2026.

These tools change frequently — check Microsoft Learn before you rely on it.

This slide deck was made with the help of AI.

It is the theme of this presentation, after all…

Learning Pathways – Adapting for AI Readiness as a Data Professional

Room 402 Chiliwack · Matt: Room 401 Chelan

- No Fluff, Just Facts: AI Essentials for Data Professionals
  Tue 10 Nov · 4:00–5:15 PM
- AI in the SQL World: Tools We Have & Tips for Using Them
  Wed 11 Nov · 10:00–11:15 AM
- Keeping Up Without Falling Down: AI's Future for Data Professionals
  Wed 11 Nov · 11:30 AM–12:45 PM



Who has been told to do X% of their work with AI?

What these tools do for us

Two help you write T-SQL. Two bring AI to your data.

- Copilot in SSMS — write and investigate T-SQL
- Copilot in VS Code — plus plan, design, custom instructions
- SQL Server 2025 — vector search in T-SQL
- SQL MCP Server + DAB — let AI agents in, safely

What are we actually solving?

What "do X% of your work with AI" looks like

- Slow query at 3 AM — Agent mode investigates
- "What does this proc do?" — Copilot explains
- Stop re-explaining your schema — save it once
- Search matches words, not meaning — vector search
- New app, no schema yet — Plan mode designs it
- AI agents need your data — SQL MCP Server governs it

And you keep control

AI runs inside the security you already have

- Copilot runs as you — your permissions, no more
- Agent mode asks before running anything
- Copilot in SSMS doesn’t keep your prompts
- You choose where the model runs — even local
- Model calls are off by default in SQL Server 2025
- Agents see only the entities you publish

GitHub Copilot in SSMS

Demo 1 · write and investigate T-SQL


Chat


Code completions


Next edit suggestions


Agent mode


Database instructions

Stop re-explaining your schema

Set it once, in the database · Database instructions · SSMS 22.3+

- Business rules stored in the database
- AGENTS.md — on a table or object
- CONSTITUTION.md — for the whole database
- Saved as extended properties
- Copilot applies them for every user

Choose who Copilot runs as

Set it once, in the database · Execution context · SSMS 22.7+

- Your login, your permissions — no more
- Approval isn't a security boundary
- agentExecuteAsUser in CONSTITUTION.md
- Point it at a low-privilege user
- No IMPERSONATE, no Copilot

Track what Copilot ran

GitHub Copilot in SSMS · pick the tool by when you need the answer

- While it runs — Extended Events
- For an auditor — SQL Server Audit
- On your machine — Copilot's log files
- In the chat — “Show me the query you used”

GitHub Copilot in VS Code

Demo 1 · plus plan, design, custom instructions


Chat (@mssql)


Agent mode


Plan mode


Schema Designer


Custom instructions

Demo 1

- SSMS: Ask mode writes it, you run it
- SSMS: Agent mode and skills find a slow query
- SSMS: instructions fix a wrong answer
- SSMS: Copilot as a low-privilege user
- SSMS: see what Copilot ran
- VS Code: instructions, Plan mode, Schema Designer

Tips for Copilot in SSMS

All five are SSMS features · VS Code uses its own instructions file

- Give it context: database instructions
- Name the server and database in prompts
- Read each step before you approve
- Avoid “Allow always” on production
- A least-privilege login for Copilot

What these tools do for us

Next: the two that bring AI to your data

- Copilot in SSMS — write and investigate T-SQL
- Copilot in VS Code — plus plan, design, custom instructions
- **SQL Server 2025 — vector search in T-SQL**
- **SQL MCP Server + DAB — let AI agents in, safely**

Words vs meaning

Keyword search matches words. Vector search matches meaning.

- LIKE '%bike%' never finds “bicycle”
- Full-text needs a thesaurus you maintain
- Vector search compares meaning
- Keep your WHERE clause — vectors add meaning

SQL Server 2025 vector search

Demo 2 · all in T-SQL


VECTOR data type


CREATE EXTERNAL MODEL


AI_GENERATE_CHUNKS


AI_GENERATE_EMBEDDINGS


CREATE VECTOR INDEX (preview)


VECTOR_SEARCH (preview)

Five steps, start to finish

Demo 2 · from model to vector search

ONCE — when you build it

Pick a model

`CREATE EXTERNAL MODEL`

↓

Embed your rows

`AI_GENERATE_EMBEDDINGS`

↓

Index the vectors

`CREATE VECTOR INDEX`

→

EVERY QUERY — for the rest of its life

Embed the question

`AI_GENERATE_EMBEDDINGS`

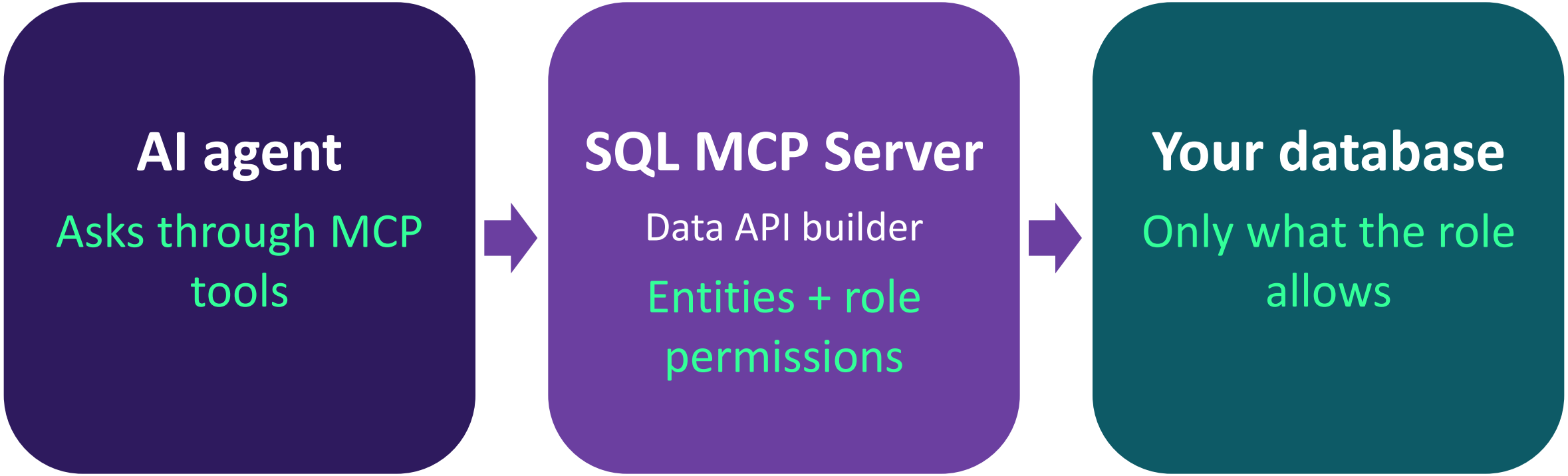
↓

Vector search

`VECTOR_SEARCH`

SQL MCP Server + DAB

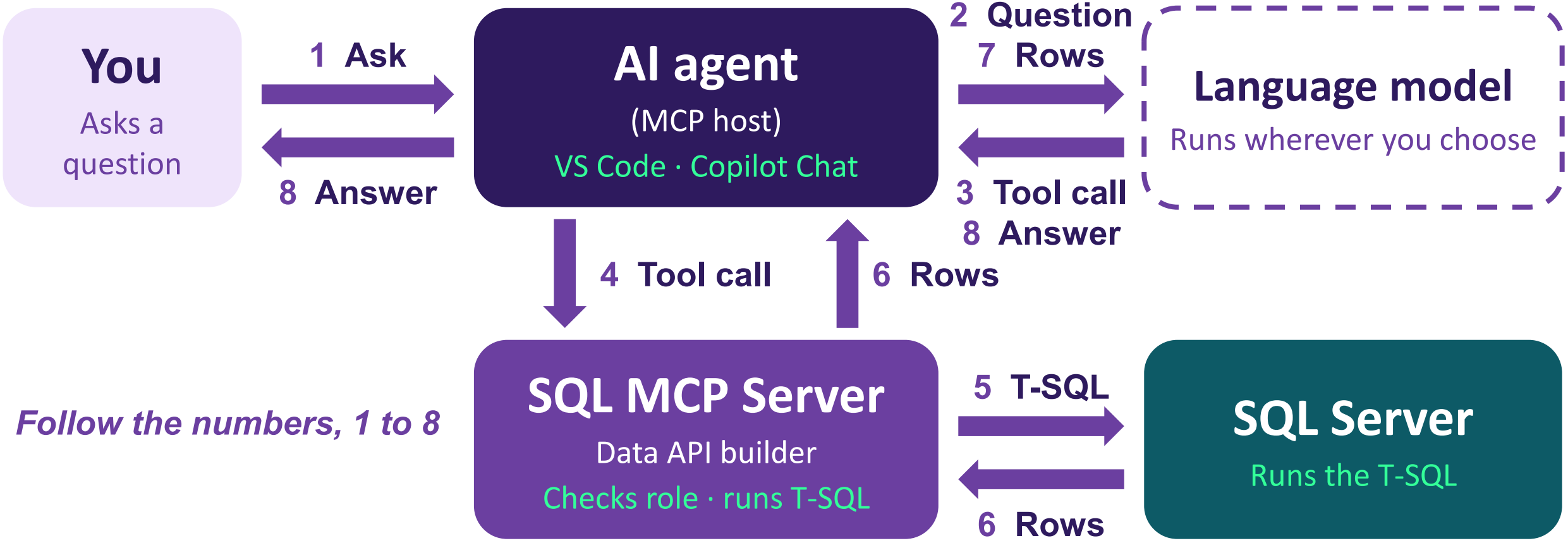
Demo 2 · agents reach data through rules you set



Open source · one config serves REST, GraphQL and MCP · DAB 1.7+, 2.0 recommended

How an agent reaches your data

Demo 2 · one question, start to finish



Follow the numbers, 1 to 8

The model picks the tool · DAB checks the role and writes the T-SQL

Demo 2

- A vector is just a column
- Search by meaning, not keywords
- Vector index + VECTOR_SEARCH
- See each model call
- MCP Server: only published entities

Tips for vector search

- Chunk long text first
- Every search sends the question to the model
- New model? Re-embed, rebuild the index
- Vector index: 100 rows minimum
- Preview on SQL Server 2025 — test first

GA or preview?

As of Oct 19, 2026 · all GA, one exception

Copilot in SSMS
Agent mode GA in 22.10.2

GA

Copilot in VS Code
MSSQL extension · agent mode, Schema Designer

GA

SQL Server 2025 — AI and vectors
VECTOR · AI_GENERATE_EMBEDDINGS · external models

GA

PREVIEW
vector index · VECTOR_SEARCH

SQL MCP Server + DAB
Data API builder 2.0

GA

The tool changed on me

Agent mode in SSMS, June to September 2026

Jun 9 · 22.7

Agent mode arrives (preview) · execution context

Jul 14 · 22.8

Built-in skills added

Aug 11 · 22.9

Ask mode stops running arbitrary queries — my demo broke

Sep 28 · 22.10.2

Agent mode GA · permissions follow the connected account

13 SSMS releases, June to October · read the release notes

Next, in Room 401 Chelan

- Matt Gordon: Keeping Up Without Falling Down
- 11:30 AM · Room 401 Chelan
- Take a short break — then head to Room 401

What to do Thursday

Start with one database

- Update SSMS and install Copilot
- Add database instructions to one database
- Give Copilot a least-privilege login
- Try vector search on one table
- Read the SQL MCP Server quickstart

Resources

Slides + scripts: github.com/sqlworldwide/Presentations

- **GitHub Copilot in SSMS**
  learn.microsoft.com/ssms/github-copilot/overview
- **GitHub Copilot in VS Code**
  learn.microsoft.com/sql/tools/visual-studio-code-extensions/github-copilot/overview
- **Vector search in SQL Server**
  learn.microsoft.com/sql/sql-server/ai/vectors
- **SQL MCP Server**
  learn.microsoft.com/sql/mcp
- **SSMS release notes**
  learn.microsoft.com/ssms/release-notes-22

Your feedback is important to us



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Thank you

Now go and try one thing on Thursday.



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